

Mixed mock exam

20 unlabeled questions · 100 practice points

Conditions: no notes, formula sheet, calculator, or answer key.

Answer style: leave answers as exact combinations, factorial ratios, products, sums, and fractions. Explain your model and what each factor or event means. Define notation you introduce.

Before every solution: write a brief method statement and the feature of the question that led you to it. There are no topic headings to supply that choice.

Suggested use: two sessions of 10 questions. As a practice option, allow about 60–75 minutes per session, then adjust based on your pace. This does not predict the real exam’s duration or format.

Each question is worth 5 self-study points: 2 for method/model, 2 for correct setup and conclusion, and 1 for explanation. The two sessions are 50 points each.

All three chapters are represented at your request. Later Chapter 3 and synthesis questions may exceed the eventual announced exam cutoff.

Name: _____

Date: _____

Session 1 time: _____ Session 2 time: _____

Prepared September 29, 2026 · Sheldon Ross, A First Course in Probability, 10th edition
Based on class notes through September 29 and homework through September 24.

SESSION 1 · QUESTIONS 1–2

X1 / 5 points

A six-character code uses the digits 0 through 9; leading zero is allowed. Count codes in which exactly one digit appears twice and each of the other four digits appears once.

Method and reason: _____

X2 / 5 points

Count integer solutions of $x + y + z + w = 16$ with $x \geq 2, y \geq 1, z \geq 0, w \geq 1$, and $z \leq 5$.

Method and reason: _____

SESSION 1 · QUESTIONS 3–4

X3 / 5 points

Suppose $P(A) = 1/2$, $P(B) = 2/5$, $P(C) = 3/10$, all three pairwise intersections have probability $1/10$, and $P(A \cap B \cap C) = 1/20$. Find the probability that exactly two of A, B, C occur, and the probability none occur.

Method and reason: _____

X4 / 5 points

Choose a committee of 6 from 14 people including A and B. The committee must include at least one of A or B. Count the committees, then count results when one committee member is designated chair.

Method and reason: _____

SESSION 1 · QUESTIONS 5–6

X5 / 5 points

On $S = \{1, 2, 3, 4\}$, singleton probabilities are $p(k) = ck$. Find c and the probability of an even outcome. Explain why this defines a probability law.

Method and reason: _____

X6 / 5 points

Nine distinct people sit at a round table. Rotations count as the same and reflections count as different. A and B must not sit next to each other. Count arrangements.

Method and reason: _____

SESSION 1 · QUESTIONS 7–8

X7 / 5 points

Two independent fair dice are rolled. Given that their sum is even, find the probability at least one die is 6.

Method and reason: _____

X8 / 5 points

A uniformly random five-card hand is dealt from a standard deck. Find the probability it contains exactly two pairs and one card of a third rank. Explain every factor.

Method and reason: _____

SESSION 1 · QUESTIONS 9–10

X9 / 5 points

Give a combinatorial proof of $r \binom{n}{r} = n \binom{n-1}{r-1}$ for $1 \leq r \leq n$.

Method and reason: _____

X10 / 5 points

A shipment comes from supplier A with probability $2/3$ and B with probability $1/3$. Their defect rates are $1/20$ and $1/5$. Given that a shipment is defective, find the probability it came from B.

Method and reason: _____

SESSION 2 · QUESTIONS 11–12

X11 / 5 points

Find the coefficient of $a^2b^3c^2$ in $(a + 2b - c)^7$. Explain your expression without calculating factorials.

Method and reason: _____

X12 / 5 points

Events $E_1 \subseteq E_2 \subseteq \dots$ satisfy $P(E_n) = 1 - 1/(n+2)$. Find the probability of their union and prove the principle you used by decomposing the events into disjoint pieces.

Method and reason: _____

SESSION 2 · QUESTIONS 13–14

X13 / 5 points

The sample space $\{1, 2, 3, 4\}$ is uniform. Let $A = \{1, 2\}$, $B = \{1, 3\}$, and $C = \{1, 4\}$. Determine whether these events are pairwise independent and whether they are mutually independent.

Method and reason: _____

X14 / 5 points

For a fixed event F with positive probability, prove $P(E^c | F) = 1 - P(E | F)$. Clearly identify the disjoint decomposition.

Method and reason: _____

SESSION 2 · QUESTIONS 15–16

X15 / 5 points

Flip a fair coin independently until the first H. Describe the finite stopped sample points, give the probability the experiment stops on the fourth flip, and discuss the event of never stopping.

Method and reason: _____

X16 / 5 points

A wins each game independently with probability $\frac{2}{3}$. In a best-of-five series, find the probability A wins the series and the probability A wins in exactly four games.

Method and reason: _____

SESSION 2 · QUESTIONS 17–18

X17 / 5 points

Uniformly select a string with 7 W's and 4 L's. Find the probability it contains exactly two maximal runs of W's.

Method and reason: _____

X18 / 5 points

A person believes $P(A) = 1/3$, $P(B) = 1/2$, $P(A \cap B) = 1/6$, and $P(A \cup B) = 5/6$. Check consistency and give the union probability required by the first three assignments.

Method and reason: _____

SESSION 2 · QUESTIONS 19–20

X19 / 5 points

Choose a fair coin or a double-headed coin, each with probability $1/2$, and toss that same coin twice. After the first toss is H, find the probability the second is H. Explain whether the tosses are independent before the coin type is known.

Method and reason: _____

X20 / 5 points

Six distinct letters are uniformly randomly placed into six addressed envelopes, one in each. Find the probability no letter is in its correct envelope, explaining how overlaps between correct-match events are handled.

Method and reason: _____
